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High-current electrical pass through standoff connection for electronic box

Abstract

A standoff connection structure which is suitable for connecting a standoff connector and a high-voltage electronic box. The standoff connection structure includes at least one anti-rotation feature, and also includes various sealing features to provide a leak-tight connection between a standoff connector and high voltage electronic box, even after exposure to thermal shock. The standoff connector also facilitates an electrical connection between a high voltage or high current component and the high voltage electronic box.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2411861	12/1945	Antony, Jr.	174/21JS	H01R 13/53

OTHER PUBLICATIONS

ISO Metric Series by Mar-Bal; ISO Series Standoff Insulators; Erico Isolators Substitue; https://www.mar-bal.com/language/en/products/standoff-insulators/mar-bal/iso-metric-series-mar-bal/ pp. 1-8; May 1, 2023. cited by applicant

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Background/Summary

FIELD OF THE INVENTION

(1) The invention relates generally to a standoff connection structure for a high-voltage box, which is leak-tight, and is able to function as a pass-through electrical connection for high current applications.

BACKGROUND OF THE INVENTION

(2) High voltage electronic boxes are generally known, and are typically used with electronic components that carry high voltage or current. Typical high voltage electronic boxes have a printed circuit board (PCB) located in a housing, which is connected to an external connector, such as a connector for a battery. However, existing designs for a connector between a battery and a high voltage box which perform as a pass-through electrical connection from the PCB to the exterior of the high voltage box are expensive to manufacture, and complicated to assemble. The connectors also do not have any type of anti-rotation feature, such that components of the connector could rotate relative to one another if ever torqued, causing damage to the connector. Current connector designs are also not leak-tight, and do not have sealing geometry to seal against the housing of the high-voltage box or prevent leakage after multiple cycles of thermal shock. The current connector designs also have exposed threads, which could present a safety concern during manufacturing assembly.

(3) Accordingly, there exists a need for a standoff connection structure which overcomes the aforementioned drawbacks, and is easily assembled.

SUMMARY OF THE INVENTION

(4) The present invention is a standoff connection structure which is suitable for connecting a standoff connector to a high-voltage electronic box. The standoff connection structure includes at least one anti-rotation feature, and also includes various sealing features to provide a leak-tight connection between a standoff connector and high voltage electronic box, even after exposure to thermal shock. The standoff connector also facilitates an electrical connection between a high

voltage or high current component and the high voltage electronic box.
(5) In an embodiment the present invention is a high voltage electronic box having a standoff connection structure, a standoff connector, a housing, a printed circuit board (PCB) disposed in the housing, and a connector recess integrally formed as part of the housing. The standoff connection structure also includes at least one anti-rotation feature which prevents the standoff connector from rotating relative to the housing. The standoff connector is connected to and in electrical communication with the PCB.
(6) In an embodiment, the standoff connector includes a main body portion, and a standoff connector housing connected to the main body portion. Part of the main body portion extends through a portion of the standoff connector housing which is disposed in the connector recess.
(7) In an embodiment, the main body portion includes a first threaded aperture integrally formed as part of the main body portion, a second threaded aperture integrally formed as part of the main body portion, and a central wall, where the central wall is disposed between the first threaded aperture and the second threaded aperture.
(8) In an embodiment, the standoff connector housing includes a flange portion and a body portion integrally formed with and adjacent the flange portion. In an embodiment, the body portion is at least partially disposed in the connector recess when the standoff connector is connected to the connector recess.
(9) In an embodiment, the connector recess includes a sidewall integrally formed as part of the high voltage electronic box, and the sidewall surrounds at least a portion of the body portion when the standoff connector is at least partially disposed in the connector recess.
(10) In an embodiment, the at least one anti-rotation feature includes a hexagonally shaped exterior surface integrally formed as part of the body portion, and the sidewall is hexagonally shaped and in contact with the hexagonally shaped exterior surface integrally formed as part of the body portion, preventing the body portion from rotating relative to the connector recess.
(11) In an embodiment, an outer seal surrounds the body portion and is adjacent to the flange portion, and an outer wall is integrally formed as part of and perpendicular to the sidewall. In an embodiment, the outer seal is located between and in contact with the flange portion and the outer wall when the standoff connector is at least partially disposed in the connector recess.
(12) In an embodiment, a groove is integrally formed as part of the standoff connector housing, and a sealant at least partially disposed in the groove and in contact with the main body portion. The sealant prevents leakage between the main body portion and the standoff connector housing.
(13) In an embodiment, the groove is integrally formed as part of the flange portion. In another embodiment, the groove is integrally formed as part of the body portion. In yet another embodiment, there are two grooves, one integrally formed as part of the flange portion, and another integrally formed as part of the body portion, and portions of the sealant are disposed in both grooves.
(14) In an embodiment, a hexagonally shaped inner surface integrally formed as part of the standoff connector housing, and a hexagonally shaped outer surface integrally formed as part of the main body portion. The hexagonally shaped inner surface integrally formed as part of the standoff connector housing is in contact with the hexagonally shaped outer surface integrally formed as part of the main body portion, preventing the main body portion from rotating relative to the body portion.
(15) Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. **1** is a perspective view of a high-voltage box having a standoff connection structure, according to embodiments of the present invention;

(3) FIG. **2** is an exploded view of a high-voltage box having a standoff connection structure, according to embodiments of the present invention;

(4) FIG. **3** is a sectional view taken along lines **3-3** of FIG. **1**;

(5) FIG. **4** is a sectional view taken along lines **4-4** of FIG. **1**;

(6) FIG. **5**A is a bottom view of a standoff connector used as part of a high-voltage box having a standoff connection structure, according to embodiments of the present invention;

(7) FIG. **5**C is a perspective view of a standoff connector used as part of a high-voltage box having a standoff connection structure, according to embodiments of the present invention; and

(8) FIG. **5**B is a is a sectional view taken along lines **5**C-**5**C of FIG. **5**A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

(10) A high-voltage box having a standoff connection structure according to an embodiment of the present invention is shown in the Figures, generally at **10**. The high-voltage box **10** having the standoff connection structure is connected to a standoff connector, shown generally at **12**. The standoff connector **12** includes a main body portion **14**, which is partially surrounded by and connected to a standoff connector housing **16**. The main body portion **14** has an exterior surface **18**, which in this embodiment is hexagonally shaped and extends the entire height **28** of the main body portion **14**. Integrally formed as part of the main body portion **14** is a first threaded aperture **20***a* and a second threaded aperture **20***b*, which are separated by a central wall **22**.

(11) The standoff connector housing **16** includes an aperture **16***a*, where the aperture **16***a* has an inner surface **24** which is hexagonally shaped to correspond to the shape of the exterior surface **18** of the main body portion **14**. The main body portion **14** is connected to the standoff connector housing **16** using a press-fit connection between the exterior surface **18** and the inner surface **24**, but it is within the scope of the invention that other types of connections, such as using an adhesive, may be used. The overall height **26** of the standoff connector housing **16** is less than the overall height **28** of the main body portion **14**, such that part of the main body portion **14** is surrounded by the standoff connector housing **16**. The standoff connector housing **16** also includes a flange portion **30***a* integrally formed with a body portion **30***b*. The difference in the heights **26**,**28** results in a first end **64** of the main body portion **14** protruding a distance **62***a* further than the flange portion **30***a* of the standoff connector housing **16**, and a second end **66** of the main body portion **14** protruding a distance **62***b* further than the body portion **30***b* of the standoff connector housing **16**. The flange portion **30***a* has an exterior surface **32** which is hexagonally shaped, and the body portion **30***b* also has an exterior surface **34** which is hexagonally shaped.

(12) The high-voltage box **10** having the standoff connection structure also includes one or more seals. Integrally formed as part of the flange portion **30***a* of the standoff connector housing **16** is at least one groove **36**, and a sealant **38***a* is disposed in the groove **36** to function as a first seal. Alternatively, or in addition to, a second groove **40** is integrally formed as part of the body portion **30***b*, and a sealant **38***b* is also disposed on the second groove **40** to function as a second seal. The sealant **38***a* and the sealant **38***b* may be the same, or may be different types of sealants, depending upon the application and the type of sealing desired. In other embodiments, the sealant **38***a*,**38***b* may be in the form of an O-ring, or other type of suitable seal. The sealant **38***a*,**38***b* is disposed in the grooves **36**,**40** and is in contact with the exterior surface **18** of the main body portion **14**, providing the sealing function between the standoff connector housing **16** and the main body

portion **14**.
(13) The high-voltage box **10** further includes a housing **42**, which may be made of metal, such as aluminum, or any other type of suitable material. The housing **42** also includes a shroud **44** which surrounds a plurality of connector pins, shown generally at **46**, where the connector pins **46** are connected to a printed circuit board (PCB) **48**, and the PCB is disposed in the housing **42**. The connector pins **46** are able to be connected to any type of suitable connector, such that the PCB **48** may be in electrical communication with a controller or other device (not shown).
(14) Integrally formed as part of the housing **42** is a connector recess, shown generally at **50**, and part of the PCB **48** is exposed in the connector recess **50**, best seen in FIGS. **3-4**. The connector recess **50** has a sidewall **52** which protrudes away from the housing **42** and is hexagonally shaped to correspond to the shape of the exterior surface **34** of the body portion **30***b*.
(15) The portion of the PCB **48** exposed in the connector recess **50** includes an aperture **48***a*. During assembly, the standoff connector **12** is positioned such that the body portion **30***b* of the standoff connector housing **16** is inserted into the connector recess **50**, the exterior surface **34** of the body portion **30***b* is in contact with the interior surface of the sidewall **52**, and the second threaded aperture **20***b* is aligned with the aperture **48***a* of the PCB **48**.
(16) A third seal, such as an outer seal **54**, which in an embodiment may be an elastomeric seal, and is shaped to correspond to the shape of an outer wall **56** of the connector recess **50**, where the outer wall **56** is perpendicular to the sidewall **52**. The outer seal **54** circumscribes the body portion **30***b* of the standoff connector housing **16** and is also adjacent the flange portion **30***a*. When the body portion **30***b* of the standoff connector housing **16** is inserted into the connector recess **50**, the outer seal **54** is positioned between the outer wall **56** of the connector recess **50** and the flange portion **30***a*. The thickness of the outer seal **54** is such that when the body portion **30***b* of the standoff connector housing **16** is initially inserted into the connector recess **50**, and prior to being secured in the connector recess **50**, the second end **66** of the main body portion **14** does not contact the PCB **48**. While in the embodiment described, the outer seal **54** is an elastomeric seal, it is within the scope of the invention that other types of seals may be used, or a sealant, such as a liquid sealant, may be used.
(17) To secure the standoff connector **12** to the connector recess **50**, a fastener **58***a*, which in this embodiment is a screw, is inserted through the aperture **48***a* of the PCB **48**, and is then inserted into the second threaded aperture **20***b*. As the fastener **58***a* is tightened, the main body portion **14** and the standoff connector housing **16** move towards the PCB **48**, and the outer seal **54** is compressed between the flange portion **30***a* and the outer wall **56**. The fastener **58***a* is tightened until the second end **66** of the main body portion **14** is in contact with the PCB **48**, securing the standoff connector **12** to the housing **42** and the PCB **48**. Because of the distance **62***b*, the standoff connector housing **16** is not in contact with the PCB **48** after the fastener **58***a* is completely tightened.
(18) Once the standoff connector **12** is secured to the connector recess **50** using the fastener **58***a*, the connection between the standoff connector **12** and the connector recess **50** is leak tight, such that the outer seal **54** prevents air or liquid from passing between the connector recess **50** and the standoff connector housing **16**, and the sealant **38***a*,**38***b* disposed in the grooves **36**,**40** prevents air or liquid from passing between the main body portion **14** and the standoff connector housing **16**. The central wall **22** also prevents air or liquid from passing between the first threaded aperture **20***a* and the second threaded aperture **20***b*. This prevents the PCB **48** from being exposed to the elements and protects the PCB **48** from corrosion and damage from exposure to debris and liquid. The outer seal **54** and the sealant **38***a*,**38***b* disposed in the grooves **36**,**40** maintain sealing after exposure to thermal shock.
(19) A ring connector **60** may be connected to the first end **64** of the main body portion **14** using a second fastener **58***b*, where the second fastener **58***b* in this embodiment is a screw. The second fastener **58***b* is inserted through an aperture of the ring connector **60** and into the first threaded aperture **20***a*, securing the ring connector **60** and a wire **62** to the main body portion **14**. The wire

62 may be connected to any type of high voltage/high current device, such as a battery. The main body portion **14** facilitates electrical communication between the PCB and the ring connector **60**, functioning as an electrical pass-through connection.
(20) The high-voltage box **10** having the standoff connection structure also includes several anti-rotation features. The contact between the exterior surface **34** of the body portion **30***b* and the sidewall **52** function as an anti-rotation feature, such that once the body portion **30***b* is inserted into the connector recess **50**, the standoff connector housing **16** and the sidewall **52** are prevented from rotating relative to one another. The contact between the exterior surface **18** of the main body portion **14** and the inner surface **24** of the aperture **16***a* also function as an anti-rotation feature, such that the main body portion **14** and the standoff connector housing **16** are prevented from rotating relative to one another. Although the exterior surface **18**, the inner surface **24**, the exterior surface **34**, and the sidewall **52** are described as being hexagonally shaped to function as the anti-rotation features, it is within the scope of the invention that the anti-rotation feature between the standoff connector housing **16** and sidewall **52**, and the anti-rotation feature between the main body portion **14** and the standoff connector housing **16** may be achieved using different shapes, such as a triangle or square. Keyway or key joint features may also be used between the standoff connector housing **16** and sidewall **52** and/or between the main body portion **14** and the standoff connector housing **16**. Furthermore, in an alternate embodiment, the main body portion **14** and the standoff connector housing **16** may be integrally formed together, such that only one anti-rotation feature between the standoff connector **12** and the sidewall **52** is needed.
(21) The description of the invention is merely exemplary in nature and, thus, variations that do not depart from the gist of the invention are intended to be within the scope of the invention. Such variations are not to be regarded as a departure from the spirit and scope of the invention.

Claims

1. An apparatus, comprising: a standoff connection structure, including: a standoff connector, the standoff connector further comprising: a main body portion; a standoff connector housing at least partially surrounding the main body portion, and the main body portion is prevented from rotating relative to the standoff connector housing by at least one anti-rotation feature; and a connector recess integrally formed as part of a housing of a high voltage electronic box; wherein the standoff connector is at least partially disposed in the connector recess such that there is a leak-tight connection between the standoff connector and the connector recess.
2. The apparatus of claim 1, the at least one anti-rotation feature further comprising: an inner surface which is hexagonally shaped and integrally formed as part of the standoff connector housing; an outer surface which is hexagonally shaped and integrally formed as part of the main body portion; wherein inner surface integrally formed as part of the standoff connector housing is in contact with the outer surface integrally formed as part of the main body portion, preventing the main body portion from rotating relative to the body portion.
3. The apparatus of claim 1, the main body portion further comprising: a first threaded aperture integrally formed as part of the main body portion; a second threaded aperture integrally formed as part of the main body portion; and a central wall, wherein the central wall is disposed between the first threaded aperture and the second threaded aperture.
4. The apparatus of claim 3, further comprising: a printed circuit board (PCB) disposed in the housing; and an aperture integrally formed as part of the PCB; wherein a fastener is inserted through the aperture and into the second threaded aperture of the of the main body portion, connecting the standoff connector to the PCB, such that the standoff connector provides electrical communication between the PCB and an electrical component located outside of the high voltage electronic box.
5. The apparatus of claim 1, the standoff connector housing further comprising: a flange portion;

and a body portion integrally formed with and adjacent to the flange portion; wherein the body portion is at least partially disposed in the connector recess when the standoff connector is connected to the connector recess.
6. The apparatus of claim 5, further comprising: an outer seal surrounding the body portion and adjacent to the flange portion; and an outer wall integrally formed as part of the connector recess; wherein the outer seal is located between and in contact with the flange portion and the outer wall when the standoff connector is at least partially disposed in the connector recess.
7. The apparatus of claim 1, the connector recess further comprising: a sidewall integrally formed as part of the high voltage electronic box; wherein the sidewall surrounds at least a portion of the standoff connector when the standoff connector is at least partially disposed in the connector recess.
8. The apparatus of claim 7, further comprising a hexagonally shaped exterior surface integrally formed as part of the standoff connector; wherein the sidewall is hexagonally shaped and in contact with the hexagonally shaped exterior surface integrally formed as part of the standoff connector, functioning as an anti-rotation feature, and preventing the standoff connector from rotating relative to the connector recess.
9. The apparatus of claim 1, further comprising at least one seal between the standoff connector housing and the main body portion.
10. The apparatus of claim 9, the at least one seal further comprising: a sealant, at least a portion of the sealant is at least partially disposed in a groove formed as part of the standoff connector housing such that the sealant is in contact with the main body portion; wherein the sealant prevents leakage between the main body portion and the standoff connector housing.
11. The apparatus of claim 10, the at least one seal further comprising a second groove integrally formed as part of the standoff connector housing, wherein another portion of the sealant is disposed in the second groove and in contact with the main body portion such that the second sealant prevents leakage between the main body portion and the standoff connector housing.
12. A high voltage electronic box having a standoff connection structure, comprising: a standoff connector, the standoff connector further comprising: a main body portion; a first threaded aperture integrally formed as part of the main body portion; a second threaded aperture integrally formed as part of the main body portion; a central wall disposed between the first threaded aperture and the second threaded aperture; and a standoff connector housing connected to the main body portion; a housing; a printed circuit board (PCB) disposed in the housing; a connector recess integrally formed as part of the housing such that part of the main body portion extends through a portion of the standoff connector housing which is disposed in the connector recess; and at least one anti-rotation feature which prevents the standoff connector from rotating relative to the housing; wherein the standoff connector is connected to and in electrical communication with the PCB.
13. The high voltage electronic box having a standoff connection structure of claim 12, further comprising: a hexagonally shaped inner surface integrally formed as part of the standoff connector housing; a hexagonally shaped outer surface integrally formed as part of the main body portion; wherein hexagonally shaped inner surface integrally formed as part of the standoff connector housing is in contact with the hexagonally shaped outer surface integrally formed as part of the main body portion, preventing the main body portion from rotating relative to the body portion.
14. The high voltage electronic box having a standoff connection structure of claim 12, the standoff connector housing further comprising: a flange portion; and a body portion integrally formed with and adjacent the flange portion; wherein the body portion is at least partially disposed in the connector recess when the standoff connector is connected to the connector recess.
15. The high voltage electronic box having a standoff connection structure of claim 14, the connector recess further comprising: a sidewall integrally formed as part of the high voltage electronic box; wherein the sidewall surrounds at least a portion of the body portion when the standoff connector is at least partially disposed in the connector recess.

16. The high voltage electronic box having a standoff connection structure of claim 15, the at least one anti-rotation feature further comprising: a hexagonally shaped exterior surface integrally formed as part of the body portion; wherein the sidewall is hexagonally shaped and in contact with the hexagonally shaped exterior surface integrally formed as part of the body portion, preventing the body portion from rotating relative to the connector recess.
17. The high voltage electronic box having a standoff connection structure of claim 15, further comprising: an outer seal surrounding the body portion and adjacent to the flange portion; and an outer wall integrally formed as part of and perpendicular to the sidewall; wherein the outer seal is located between and in contact with the flange portion and the outer wall when the standoff connector is at least partially disposed in the connector recess.
18. The high voltage electronic box having a standoff connection structure of claim 14, further comprising: a groove integrally formed as part of the standoff connector housing; and a sealant at least partially disposed in the groove and in contact with the main body portion; wherein the sealant prevents leakage between the main body portion and the standoff connector housing.
19. The high voltage electronic box having a standoff connection structure of claim 18, wherein the groove is integrally formed as part of the flange portion.
20. The high voltage electronic box having a standoff connection structure of claim 18, wherein the groove is integrally formed as part of the body portion.